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The content extracted is as follows:

The vascular endothelium, which plays an important role in organ homeostasis, is a key target of radiation damage. Microvascular endothelial cells (EC), in particular, are sensitive to radiation and radiation-induced alterations in EC structure and function.

Radiation-induced endothelial damage and its associated vascular changes often lead to chronic lesions when organs at risk (e.g. lung, kidney, heart and brain) are exposed to sufficiently high doses [3], and patients with IR-induced tissue damage may die of organ failure.

Given the central role of the endothelium in normal vascular function, a better understanding of mechanisms of radiation-induced dysfunction is vital in developing therapeutics for radiation-induced endothelium damage [3].

Endothelial dysfunction in irradiated tissue

Radiation-induced endothelium injury can manifest acutely/ early or late after IR exposure.

IR-induced cellular damage triggers apoptosis and cell death leading to the release of damage-associated molecular patterns (DAMP) and activation of the systemic inflammatory response producing significant changes in the microvascular network structure and function [29–32].

A key step in IR-induced organ damage is excessive adhesion to and migration of activated leukocytes across vasculature (Supplementary Data Video 1 and Video 2) and a reduction in neutrophil infiltration is associated with better outcomes following skin irradiation [4].

A number of molecules involved in the leukocyte adhesion cascade are also involved in radiation-induced tissue damage.

Ability of peripheral blood neutrophils from cervical cancer patients to form NETs increased from 53.6% before radiation therapy to 66.7% after radiation therapy [56] but the mechanisms by which IR activates NETs and the interconnection between NETs and radiation-induced EC damage have not been fully delineated. Damage to and/or the denudation of EC after radiation exposure produces changes in permeability of EC.

Endothelial cell apoptosis at doses > 5 Gy can be induced by persistent DNA damage, resulting in leading to p53 accumulation and resulting in activation of the caspase pathway, [72] but mechanisms of endothelial cytotoxicity at lower radiation doses are less known [73].

Extracellular acidic sphingomyelinase then leads to radiation-induced, ceramide-mediated EC apoptosis [75].

Although no single animal model can completely represent the human condition, in vivo models including mouse, rat, rabbits, and pigs have been widely used to study radiation-induced endothelium damage [68].

The choice of an appropriate animal model for a specific study is generally dictated by factors such as anatomy, physiology, genetics, and immune response of the animal model being considered. Mice show similarities to human physiological and genome (>95%) [79] and have been used widely to study radiation induced upregulation of cell adhesion molecules, leukocyte endothelial interaction and EC damage [34,80].

However, the mouse may not be the optimum animal model for studying radiation induced damage for multiple reasons.

Given the complexities inherent in the use of animal models, several in vitro systems have been developed that use human cell-based assays to provide a better understanding of radiation-induced cell damage in humans.

These new models not only provide novel approaches for studying radiation damage to EC but also offer a new tool for the rapid development and screening of therapeutics.

Although organoids have in vivo like cellular architecture and spheroids have advantages of high reproducibility [92], they are still not widely used in studying radiation-induced EC damage in part due to the major challenges associated with developing organoids and spheroids.

For example, a 3D capillary model using human umbilical vein EC (HUVEC) grown in a collagen gels showed more complex and persistent 53BP1 DNA damage foci when exposed to high LET (linear energy transfer) as compared to low LET [97].

For example, Gut-on-a-chip models have been used to study the effects of gamma (γ)-radiation on villus intestinal epithelium [88].

Importantly, a systematic comparison between HUVEC cultured in this system with a traditional 2D monolayer model showed significant differences upon IR exposure, particularly at high doses that are

typically used in stereotactic body radiation therapy (SBRT) and stereotactic radiosurgery (SRS).

At high IR doses up to 25 Gy, VE-cadherin cell cell adherens junctions did not change significantly in the 3D microfluidic model but were significantly damaged in the 2D monolayer model.

Therapeutics

While the use of Neupogen and Neulasta to treat hematopoietic acute radiation syndrome (H-ARS) was recently approved by the FDA, therapeutic approaches to the treatment of radiation-induced vascular/endothelial injury are largely supportive and there are no specific pharmacologic therapies available that protect from radiation-mediated tissue damage [3].

This lack of effective anti-inflammatory agents continues to complicate the treatment of inflammatory pathologies ranging from sepsis and COVID-19 to radiation-induced inflammatory response.

Radiotherapies for treatment of IR-induced endothelial damage have been primarily focused in 3 areas, targeting EC pathways activated in response to radiation, reconstitution of bone marrow and reduction in vascular barrier permeability and vascular leak [3].

Several potential therapeutics that target IR-induced damage to EC are currently being evaluated in in vitro cell cultures and animal models.

In vivo studies demonstrated that Ac-SDKP administration was able to restore IR-induced endothelial barrier damage by preventing EC cell loss and increasing expression of tight junction proteins.

Although the mechanisms of TM transcription activation by this drug is unclear, suppression of IR-induced endothelial interaction with leukocyte may prevent intestinal inflammation post-irradiation [107].

Genetic approaches employing knockout mice have identified other EC targets to mitigate IR-induced cell damage.

PKCδ is activated in multiple cell types in response to radiation exposure and is involved in radiation-induced apoptosis [113].

Moreover, PKCδ overexpression enhanced radiation-induced apoptosis indicating a critical signaling role [114]. PKCδ^{-/-} mice were protected from

radiation-induced damage to the salivary gland and thymus [115].

Similarly, we have used our novel microfluidic assay (bMFA) to show that PKCδ inhibition dramatically attenuates radiation-induced human EC damage, leukocyte migration (Fig. 4A) as well as vascular EC permeability (Fig. 4B) and preserved integrity of irradiated EC (Fig.

PKCδ inhibition downregulated radiation-induced P-selectin, ICAM-1 and VCAM-1 overexpression [41].

In summary, radiation-induced EC injury in large part mediates and regulates wider tissue damage.

Emerging in vivo and in vitro models of vascular inflammatory response are providing not only a better understanding of mechanisms underlying the progression of radiation-induced EC damage but also a roadmap for developing highly specific radiotherapeutics for preventing and treating side effects of radiotherapy and/or accidental radiation exposure.

Novel therapeutics that specifically focus on common key control points or signaling hubs can more effectively regulate the multiple overlapping and redundant mechanisms that modulate the signaling pathways regulating the vascular endothelial response to radiation-induced damage.